Knowledge-Based Systems 258 (2022) 110036

Contents lists available at ScienceDirect

Knowledge-Based Systems

journal homepage: www.elsevier.com/locate/knosys

A survey on deep learning based knowledge tracing

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ARTICLE INFO

Article history:
Received 13 September 2022
Received in revised form 12 October 2022
Accepted 14 October 2022
Available online 23 October 2022

Keywords:
Knowledge Tracing
Deep learning
Educational data mining
Intelligent tutoring systems
Graph neural network

ABSTRACT

"Knowledge tracing (KT)" is an emerging and popular research topic in the field of online education that seeks to assess students' mastery of a concept based on their historical learning of relevant exercises on an online education system in order to make the most accurate prediction of student performance. Since there have been a large number of KT models, we attempt to systematically investigate, compare and discuss different aspects of KT models to find out the differences between these models in order to better assist researchers in this field. The findings of this study have made substantial contributions to the progress of online education, which is especially relevant in light of the current global pandemic. As a result of the current expansion of deep learning methods over the last decade, researchers have been tempted to include deep learning strategies into KT research with astounding results. In this paper, we evaluate current research on deep learning-based KT in the main categories listed below. In particular, we explore (1) a granular categorisation of the technological solutions presented by the mainstream Deep Learning-based KT Models. (2) a detailed analysis of techniques to KT, with a special emphasis on Deep Learning-based KT Models. (3) an analysis of the technological solutions and major improvement presented by Deep Learning-based KT models. In conclusion, we discuss possible future research directions in the field of Deep Learning-based KT.
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1. Introduction

In recent years, millions of individuals have chosen to study via online platforms as a result of the expansion and popularity of online education platforms such as Intelligent Tutoring Systems (ITS) [1] and Massive Open Online Courses (MOOCs) [2]. The most significant advantage of online learning systems over conventional offline education is that they maintain complete student trajectories, enabling the evaluation of the effectiveness and knowledge mastery of student behaviour over several learning sessions. On online learning systems, the imbalance between the number of students and instructors renders manual tutoring unfeasible. Academics have paid close attention to the topic of how to employ online learning systems to anticipate potential learning patterns from students' learning trajectories and provide personalised assistance for the purpose of Artificial Intelligent (AI)-assisted education.

Knowledge Tracing (KT), is an effective instrument for AI-assisted education. In recent years, it has also developed into an important component of ITS, which is used extensively in many different online education platforms, like Coursera [3], edX [4], and ASSISTment [5]. The goal of KT is to model students' mastery of a concept based on their historical learning of relevant exercises. Given a student's historical interactions with various learning exercises $\{X_0, \ldots, X_t\}$, the KT task may often be formalised as a supervised sequential learning task. Using a pre-defined model from which to retrieve the student's implicit knowledge state monitored over time, the learning interaction is often shown as a exercise-answer tuple, denoted by the notation $X_t = \{e_t, a_t\}$. This notation shows that the student provides an answer to the exercise denoted by e_t at the instant denoted by t, and that the response is denoted by a_t. Because it might be difficult to accurately quantify the actual learning state of the learner, current KT models often make use of an alternate approach that enables the model to predict the student's probability of correctly responding to the next exercise $P_{X_{t+1}}(a_{t+1} = 1 \mid e_{t+1}, X_t)$.

As shown in Fig. 1, this is a simple schematic representation of a KT model. As a student interacts with the platform, the learning platform continuously records data about the student's interactions, such as exercises, the correctness of the exercises and the concepts (line graphs, probabilities, linear equations, etc.) that correspond to these exercises. Based on this recorded interaction data, KT aims to keep a continuous trace of all the

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https://doi.org/10.1016/j.knosys.2022.110036
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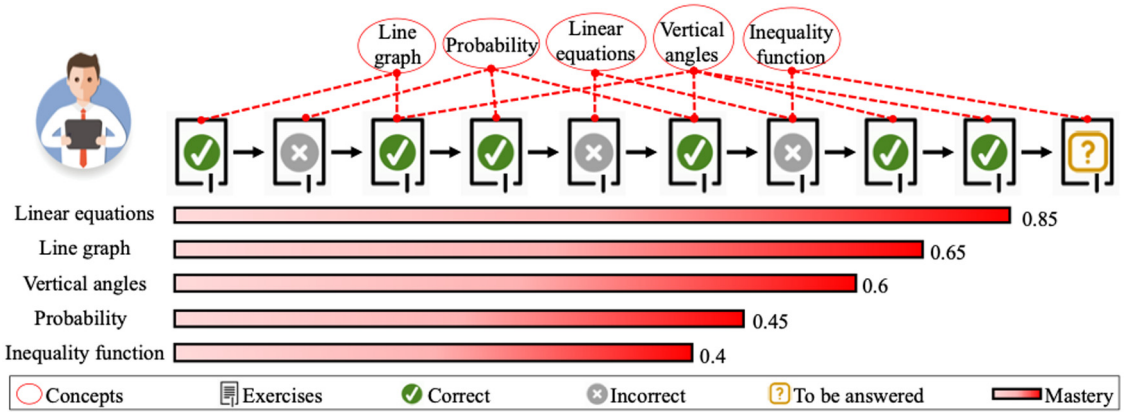

Fig. 1. An overview of Knowledge Tracing.

concepts covered by the student in order to intervene with due diligence and in a timely manner when intervention is needed. In Fig. 1, for example, the student's latest mastery rates for the five concepts are 0.85, 0.65, 0.6, 0.45 and 0.4, which also represent the probability that the student will get a new exercise right for the corresponding concept. The KT model is designed to monitor the evolution of students' knowledge in order to provide them with intelligent services, such as recommending exercises that are appropriate to their level of ability or alerting educators to manual interventions. In fact, KT is of great interest to both the educator side and the learner side. Not only does it give students a better understanding of their own learning, it also allows the platform to tailor exercises to each student in order to improve their academic performance faster and more accurately [4,6–19].

Bayesian Knowledge Tracing (BKT) [20], one of the most well-known KT models, was the first model to employ a Hidden Markov Describe (HMM) to model a student's prospective information state as a collection of binary variables that indicate whether a concept or body of knowledge c has been learned. In a broader sense, concepts are often referred to as knowledge concepts, principles, facts, or abilities. The knowledge required to answer a particular exercise properly is a notion c. The majority of existing models consider e_t and c to be the same thing. Since its introduction 20 years ago, BKT has been regarded as the most effective model in the area of KT. Throughout this period, several machine learning models have been developed in an attempt to tackle the KT issue. These models include several BKT variants [21], logistic regression [22], and item response theory (IRT) [23]. Numerous individuals have attempted these cutting-edge techniques from the same time period, but they do not perform significantly better than BKT [24]. Even though KT has achieved great success, it still faces significant challenges. First, the correspondence between variables and concepts is not one-to-one, is ambiguous, and the binary variables are not set up in a way that is consistent with the reality of the learning process [25]. Furthermore, the way in which these variables were set up was not consistent with the fact that they were binary. Secondly, when modelling each concept independently, BKT fails to capture the inter-concept relationships and loses the ability to model both the final defined concept and the complex concept at the same time. Therefore, attempts have been made to try and find reasonable ways to address these issues.

As the rise of deep learning in the recent decade [26], deep learning has also been attempted by many researchers to be applied to the KT domain due to its excellent feature extraction capabilities. This application of deep learning in the KT domain is known as Deep Learning-based KT. Compared to typical machine learning models, Deep Learning-based KT does not require manually annotated conceptual information. Furthermore, it is able to capture more complex representations of student knowledge, as well as uncover and exploit information about the relationships between concepts. Deep Learning-based KT has now grown to become one of the most important research centres in the KT field.

In 2015, the Deep Knowledge Tracing (DKT) [25] model was introduced, which is considered to be the foundational work of Deep Learning-based KT. However, although the predictive performance of DKT outperforms existing classical methods, it has been criticised by some researchers for its utility in educational applications [27,28]. This is mainly because of the following three reasons: (1) the hidden states $\boldsymbol{h}_t$ are inherently difficult to interpret as knowledge states, and the DKT model does not deeply analyse knowledge interactions [29], making it poorly interpretable. (2) RNN has a long-term dependency problem, and its variants (e.g. LSTM and GRU) only increase the capacity of sequence learning [30–32], but do not solve the problem. (3) In DKT, the model uses a single coded sequence of student interactions, using only the interaction sequence as a feeder, wasting the rich learning trajectory information retained by the online platform, and the performance of the model is affected by the small number of learning features in the feeder. Poor interpretability, long-term dependency and few learning features are the three most significant problems with the DKT model, and many researchers have worked to extend and improve it to address these issues [29,32–46]. These models address these problems using different embedding approaches, the design of loss functions, the creation of new data structures, and the introduction of attention mechanisms, respectively. As these models continue to be proposed, a survey based on these newly proposed models is urgently needed to be written to summarise them.

It is important to note that while there have been several attempts to survey or benchmark the entire KT model [47–49] or deep learning-based KT models [10,50–52], the former have generally focused mainly on summarising the development of the entire KT field and illustrating the relevant background concepts, while the latter tend to presented only a few representative models lacking systematic analysis and comparison. Both of them do not deeply examine and analyse the unique contributions of different deep learning-based KT models and their improvement directions. We advocate that we focus on the field of deep learning-based KT and conduct an in-depth survey of the technical developments as well as the latest research advances in the field. The main contributions are listed below.

- As we have covered in this work, especially in recent years, there has been a large amount of research on the particular problem of deep learning-based KT. We propose an explicit taxonomy of categories for deep learning-based KT methods

and compare their architectures in several aspects, including model design, knowledge state representation, assumptions about the relationships between exercises and concepts, and improvement issues.
- We provide an in-depth evaluation and summary of these models using four real-world KT datasets, and compare the performance of key deep learning-based KT methods on each dataset by integrating results from the relevant literature.
- We provide important insights into the deep learning-based KT problem as well as discuss several issues and limitations of deep learning-based KT that are currently not well explored to help derive future research directions in new venues.

The rest of the paper is organised as follows. In Section 2, we first review the problem definition of the KT problem and the generic KT paradigm. Next, we provide a practical classification and analysis of the deep learning-based KT models in our survey in Section 3, while we summarise each model in detail. In Section 4, we present the benchmark dataset we prepared and analyse each individual model in detail and conduct a comparison and discussion of the experimental results. Finally, we conclude our study in Section 5 and give some interesting insights into the current deep learning-based KT models and possible future research directions.

2. Overview of knowledge tracing

In this section, we present an overall introduction to KT to better understand the necessary background and preliminaries. We give a brief introduction to them in two different aspects, including problem definition and Common paradigm.

2.1. Problem definition

From 1994, when the concept of KT was first introduced by [20], to 2015, when deep learning models were first applied to the problem of KT [25], until now dozens of KT models based on deep learning models [34–36,53] have been proposed one after another, its problem definition has remained largely consistent. It can be summarised as follows, there are three main elements in an intelligent education system, the student $|S|$, the exercise (question) $|E|$ and the knowledge concept (skill) $|C|$ corresponding to the exercise, and the interaction $|X|$ among these three elements is the main activity in an intelligent education system.

Problem 1 (*Knowledge Tracing*)**.** Given a student's historical exercise interactions $s \in |S|, s = \{X_0, \ldots, X_t\} = \{(e_0, a_0), (e_1, a_1), \ldots, (e_t, a_t)\}$ on a specific concept $c \in |C|$, where $X_t = (e_t, a_t)$ for the exercise $e \in |E|$ is a tuple that denotes the correctness a_t of the result obtained at step $t \in T$, where $a_t \in \{0, 1\}$ is a binary that equals 1 if correct and 0 if incorrect. The knowledge tracing task is to predict the next interaction X_{t+1}.

2.2. Common paradigm

In the KT tasks, researchers have tended to embed student learning processes through different models and methods. However, similar settings are adopted for the training targets under different models and methods, i.e. the negative log-likelihood of the observed sequence of student responses. Thus, the final training objective can usually be generalised to a mathematical paradigm that incorporates most present and possibly many future KT models and methods.

The objective function of the KT task is to learn the negative log likelihood of a given student's sequence of practice procedures from step 1 to T through observation of the student's previous interactions. Formally, at step t, let $\mathcal{L}$ be the binary cross-entropy, let $\tilde{a}_t$ be the predicted score for exercise e_t through the model, and a_t be the actual binary score, so that the overall loss $\mathcal{L}$ for a given student is defined as.

$$\mathcal{L} = -\sum_{t=1}^{T} \left(a_t \log \tilde{a}_t + (1 - a_t) \log \left(1 - \tilde{a}_t\right)\right) \tag{1}$$

3. Model categorisation of deep learning-based KT

The deep learning-based KT models are summarised in Table 1. Specifically, we classify these models according to the main techniques as DKT and its Variants (RNN based Deep Learning-based KT), Memory Network based Deep Learning-based KT, Attention Mechanism based Deep Learning-based and Graph Structural Deep Learning-based KT models. We also summarise the three main improvement issues according to DKT and the most major contribution of each individual model. More details will be explained in this section.

3.1. DKT and its variants

Deep Knowledge Tracing (DKT) [25] pioneers the use of deep learning for knowledge tracing. It employs Recurrent Neural Network (RNN) to predicate the probability of correctly answering an exercise at each time step. A sequence of hidden states $\langle h_1, h_2, \ldots, h_n \rangle$ is computed which encodes the sequence information obtained from previous interactions. Despite the great success of DKT in the field of KT, it still faces many problems and limitations. In order to improve DKT, a number of variant (extended) models based on DKT have been proposed, and in this section we focus on DKT, the originator of Deep Learning-based KT, and its variant models.

3.1.1. DKT

The DKT model proposed by Piech et al. [25] is a pioneering work in the field of Deep Learning-based KT and a fundamental model in the field of Deep Learning-based KT. The RNN is a mnemonic sequential model with a sequential structure that conforms to the proximate causality effect in learning and preserves the learning trajectory information. This property makes variants of RNN (including the long short term memory network (LSTM) and the gated recurrent network (gated recurrent unit, GRU) the most widely used models in the Deep Learning-based KT field. DKT takes student learning interactions $\{x_1, x_2, \ldots, x_t\}$ as the input, and through one-hot encoding or compressed sensing (compress sensing), x_t is transformed into an input vector. In DKT, the hidden state $\boldsymbol{h}_t$ of RNN is interpreted as the student's knowledge state, and $\boldsymbol{h}_t$ is further predicted by a Sigmoid-activated linear layer $\boldsymbol{y}_t, \boldsymbol{y}_t$ is equal in length to the number of exercises, each element of which represents the predicted probability of the student answering the corresponding exercise correctly. This is calculated as

$$\begin{aligned} \boldsymbol{h}_t &= \tanh\left(\boldsymbol{W}_{hx}\boldsymbol{x}_t + \boldsymbol{W}_{hh}\boldsymbol{h}_{t-1} + \boldsymbol{b}_h\right), \\ \boldsymbol{y}_t &= \text{Sigmoid}\left(\boldsymbol{W}_{yh}\boldsymbol{h}_t + \boldsymbol{b}_y\right) \end{aligned} \tag{2}$$

Training is aimed at maximising the negative log probability of the observed sequence of student answers within the framework of the model. Let $\delta(e_{t+1})$ be the one-hot encoding of the exercise whose response has been received at time $t+1$, and let ℓ denote the binary cross entropy. For each given prediction, the loss

Table 1
Classification summary of existing studies.

Model	Main technique category	Major improvement issues			Major contribution
		Long-term dependency	Interpretability	Side information	
DKT	DKT and its Variants (RNN based Deep Learning-based KT)	–	–	–	1st model introduces deep learning in KT
DKT+		✓			Improved time-series self-attention calculation
DKT-DSC		✓		✓	Training students by ability and implicit features
EERNN			✓	✓	Designed a BiLSTM to extract exercise semantic representations from texts
DKVMN	Memory Network based Deep Learning-based KT		✓		Introduce the attention between exercise and concept
SKVMN		✓	✓		Construct Hop-LSTM and topological relations
SAKT	Attention Mechanism based Deep Learning-based KT	✓			Introduction of transformer encoder structure
AKT				✓	Extraction of guess and miss behaviour features
SAINT		✓			Complete Transformer migration for KT
SAINT+		✓			Extend Transformer model for migration applications
RKT				✓	Addition of a relation-aware contextual self-attention layer
GKT	Graph Structural Deep Learning-based KT models			✓	Adding topic topology relationships using graph neural networks
GIKT				✓	Introduction of graph convolutional networks in KT
PEBG			✓	✓	Conduct pre-training embeddings for exercises with rich side information
JKT			✓	✓	Establishing e2e and c2c relationships as additional exercise features
Bi-CLKT			✓	✓	Applying contrastive learning to enhance e2e and c2c relationships embeddings

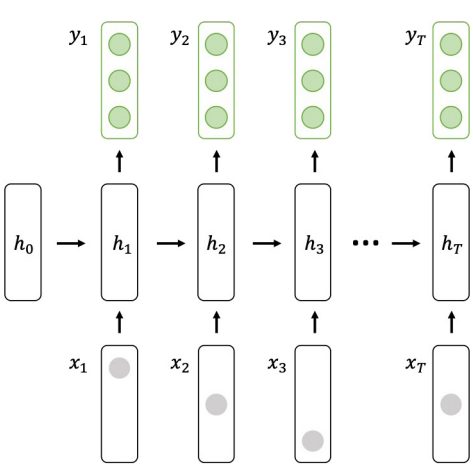

Fig. 2. An overview of DKT.

is written as $\ell\left(\mathbf{y}_i^T \delta\left(e_{t+1}\right), a_{t+1}\right)$, whereas the loss for a single student is written as:

$$L = \sum_t \ell\left(\mathbf{y}_i^T \delta\left(e_{t+1}\right), a_{t+1}\right) \tag{3}$$

For learning-centred teaching and learning assessment systems, DKT is preferable to the traditional machine learning model represented by BKT because it outperforms BKT without the need for human annotated data (AUC increased by 20 percent) and can capture and exploit deeper representations of student knowledge (see Fig. 2).

3.1.2. The variants of DKT

DKT's strong performance has its limits. First, it presupposes that $\boldsymbol{h}_t$ conceals a single hidden concept. Second, numerous inter-concept connections cannot be simulated. Third, it presupposes that all exercises have an equal chance of being connected, which may not be true in many circumstances. Some exercises in the sequence may be more vital. To overcome its drawbacks, the DKT model has been extended.

Extended-Deep Knowledge Tracing is proposed by Xiong et al. [54] to broaden DKT by introducing aspects such as textual information, exercise difficulty, concept level, and concept reliance in students' practice sessions.

DKT+ [14] augments the basic DKT loss function with two extra regularisation terms to overcome DKT's limitations in reconstructing response inputs and reducing inconsistency in predicting replies to exercises with comparable concept.

Minn et al. [55] present DKT-DSC, which utilises K-means to cluster student profiles based on concept performance and dynamically updates clustering information as performance changes. To do this, they first divide the interaction sequence into discrete time intervals, then compute the student's right and erroneous responses for each concept at each interval, and then use the difference between these two values as a measure of the student's proficiency:

$$\begin{aligned} \text{correct}\left(x_j^z\right) &= \sum_{i=1}^{\varepsilon} \frac{q_j == 1}{N_{i,i}}, \\ \text{incorrect}\left(x_j^z\right) &= \sum_{i=1}^{\varepsilon} \frac{q_j == 0}{N_{i,t}}, \\ r\left(x_j^z\right) &= \text{correct}\left(x_j^z\right) - \text{incorrect}\left(x_i^z\right), \\ d_i^z &= \left(r\left(x_i^z\right), r\left(x_2^z\right), \ldots, r\left(x_n^z\right)\right). \end{aligned} \tag{4}$$

where z denotes the time interval and $N_{j,i}$ denotes the number of times q_j has been answered in the previous i time interval. The $\boldsymbol{d}_i^z$ is interpreted as the learning ability of student i. Then, students are grouped by ability using the K-means (K-means) algorithm:

$$\text{cluster}(i, z) = \underset{c \in [1,m]}{\arg\min} \sum_{i=1}^{m} \sum_{d_i^{z-1} \in C_i} \left\| \boldsymbol{d}_i^{\varepsilon-1} - \boldsymbol{\mu}_i \right\|^2.$$

Finally, the student data were placed in different groups to train the model.

3.2. Memory network based deep learning-based KT

Several efforts, inspired by memory-augmented neural networks [53], have improved DKT by supplementing an external memory structure to better track complicated concepts learnt by students. In particular, a key–value memory has been utilised to represent knowledge state, in line with Key-Value Memory Network (KVMN) [56], which offers greater representational capacity than a hidden variable does in DKT. A key–value memory has two matrices: one for the key, and another for the value. The key matrix holds the concept representations, while the value matrix stores the student's degree of competence with respect to each concept. Here, we compare and contrast two widely used key–value memory networks for tracing past experiences.

3.2.1. Dynamic Key-Value Memory Network (DKVMN)

As Shown in Fig. 3, DKVMN (dynamic key–value memory network) [35] is inspired by MANN (memory augmented neuralnetwork). DKVMN uses the matrix $\boldsymbol{M}^{\mathrm{k}}$ to store concept, each column of $\boldsymbol{M}^{\mathrm{k}}$ represents a concept; the matrix $\boldsymbol{M}^{v}$ is used to store the knowledge state, each column of $\boldsymbol{M}^{v}$ to store the knowledge state, each column of $\boldsymbol{M}^{v}$ represents the mastery level of a concept in $\boldsymbol{M}^{\mathrm{k}}$. The attention mechanism is mainly reflected in the computation of w_i: at moment t, the embedded vector $\boldsymbol{e}$ is inner-products with each concept (i.e. each column of $\boldsymbol{M}^{\mathrm{k}}$), and after an activation w_i, $w_i(i)$ represents the correlation between the topic q_t and $conceptc_i$, i.e. the attention weights:

$$w_t(i) = \text{Softmax}\left(\boldsymbol{e}_t \boldsymbol{M}_t^{\mathrm{k}}(i)\right). \tag{5}$$

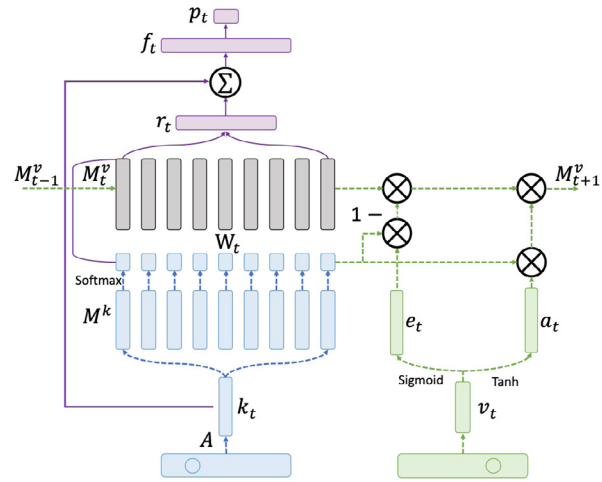

Fig. 3. An overview of DKVMN.

Calculate the student's grip on the question q_t based on the attention weights $\boldsymbol{r}_t$:

$$\boldsymbol{r}_t = \sum_i w_t(i) \boldsymbol{M}_t^v(i). \tag{6}$$

The exercise's embedded vector $\boldsymbol{e}_i$ implicitly contains difficulty information, and by concatenating it with r_t, we obtain f_t which contains both difficulty and mastery information.

$$\boldsymbol{f}_t = \tanh\left(\boldsymbol{W}_1^{\mathrm{T}}\left[\boldsymbol{r}_t \oplus \boldsymbol{e}_t\right] + \boldsymbol{b}_1\right). \tag{7}$$

Finally, $\boldsymbol{f}_t$ obtains a prediction of student performance via a Sigmoid-activated linear layer $\boldsymbol{y}_t$:

$$\boldsymbol{y}_t = \text{Sigmoid}\left(\boldsymbol{W}_2^{\top} \boldsymbol{f}_t + \boldsymbol{b}_2\right). \tag{8}$$

3.2.2. Sequential Key-Value Memory Network (SKVMN)

Abdelrahman et al. [57] use the triangular membership function in the SKVMN model to compute the correlation between concepts, which is different from the standard LSTM. SKVMN was developed to overcome a shortcoming of DKT and DKVMN, namely, that concepts needed to answer previous exercises in a sequence are not necessarily relevant to concepts needed to answer the current exercise. With the same key–value memory structure and loss function as DKVMN, SKVMN uses a modified LSTM for sequential modelling. This variant is known as Hop-LSTM.

$$\mu(x) = \max\left(\min\left(\frac{x-a}{b-a}, \frac{c-x}{c-b}\right), 0\right), \tag{9}$$

where a, c define the triangle's "legs", and b its "peaks". As can be seen in Fig. 4, correlation is employed as a constraint on the Hop-LSTM to regulate the hopping of information on subjects and as a means of further computing the sequential relationships between them.

3.3. Attention mechanism based deep learning-based KT

Since many researchers have criticised the interpretability of DKT, many approaches have been tried to improve this problem. Ultimately, it is found that this could be done by developing learning models that embed interpretability directly into the structure of individual models. The inherent interpretability of the models may be accomplished by constructing learning models that directly include interpretability into the structure of particular models to boost the interpretability of Deep Learning-based KT models.

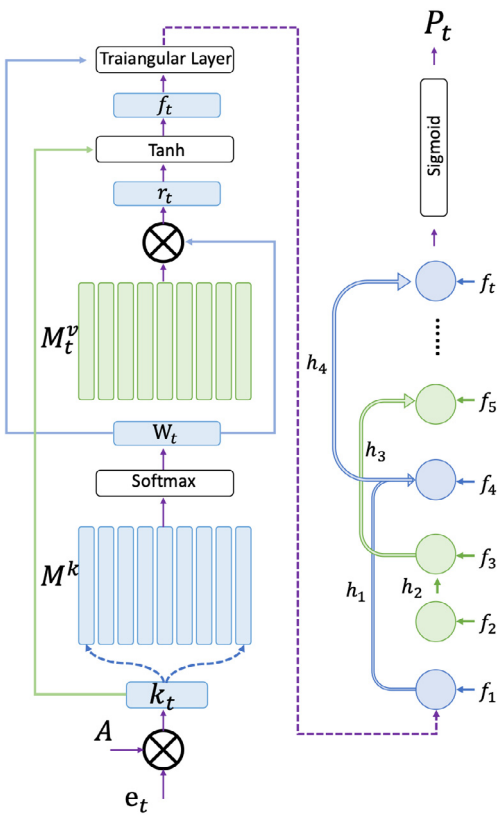

Fig. 4. An overview of SKVMN.

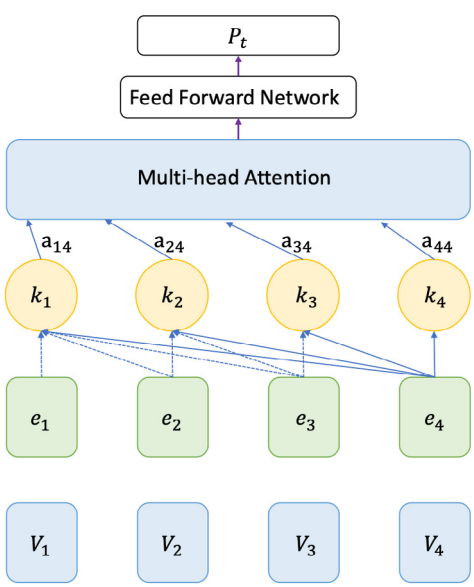

Fig. 5. An overview of SAKT.

Using the Transformer structure, many researches [29,34,58–60] have attempted to include an attention mechanism in KT tasks. While each of these works introduces a slightly different mechanism for focusing attention, they all share a common objective: to learn the attention weights of exercises in an interaction sequence in a manner that accurately reflects their relative importance when attempting to predict the probability of answering the next exercise correctly. This solves a shortcoming of DKT by decreasing its inclination to give each exercise in a series of exchanges identical weight.

3.3.1. Self-Attentive Knowledge Tracing (SAKT)

Pandey et al. [34] first apply the usage of Transformer models in the domain of knowledge tracking by introducing the SAKT model, whose architecture is depicted in Fig. 5.

In Transformer, the three attention-calculating parameters $\boldsymbol{E}$, $\boldsymbol{K}$, $\boldsymbol{V}$ are obtained by multiplying the input sequence by various weight matrices. In SAKT, E and K, V are obtained by projecting, respectively, the embedded sequences of the topic and the interactive embedded sequences.

$$E = \hat{E}W^2, K = \hat{X}W^K, V = \hat{X}W^v, \tag{10}$$

where $\hat{E}$ is the topic embedding matrix and $\hat{\boldsymbol{X}} = \boldsymbol{X} \oplus \boldsymbol{P}$ is the topic embedding matrix derived by concatenating the interaction security matrix $\boldsymbol{X}$ with the position embedding matrix $\boldsymbol{P}$. SAKT is not a temporal model. Hence, position embedding is required to maintain the position information of the input sequence. Attention is computed h times so that the model can learn relevant information in multiple representation subspaces, and the results are concatenated; this is referred to as multi-head attention [61].

$$\begin{aligned} \boldsymbol{H}_{\text{MHA}} &= [\boldsymbol{H}_1 \oplus \boldsymbol{H}_2 \oplus \cdots \oplus \boldsymbol{H}_h]\, \boldsymbol{W}^{\text{MHA}}, \\ \boldsymbol{H}_i &= \text{Attention}\ (\boldsymbol{Q}_i, \boldsymbol{K}_i, \boldsymbol{V}_i) = \text{Softmax}\left(\frac{Q_i \boldsymbol{K}_i^\top}{\sqrt{d}}\right) \boldsymbol{V}_i, \end{aligned} \tag{11}$$

where d represents the size of the embedded vector. Using a prebarium nerve network layer and a Sigmoid-activated linear layer, the following predictions were obtained:

$$\begin{aligned} \boldsymbol{F} &= \text{ReLU}\,(\boldsymbol{H}_{\text{MHA}}\boldsymbol{W}_1 + \boldsymbol{b}_1)\,\boldsymbol{W}_2 + \boldsymbol{b}_2, \\ \boldsymbol{y}_t &= \text{Sigmoid}\,(\boldsymbol{F}_1\boldsymbol{W} + \boldsymbol{b})\,. \end{aligned} \tag{12}$$

3.3.2. Context-aware attentive KT (AKT)

AKT [29] contextualises raw embedding by evaluating the student's whole interaction history and by employing a context-aware depiction of historical exercises and responses. AKT provides a novel monotonic attention method that employs an exponential decay curve to diminish the significance of exercises from the remote history. This mechanism was inspired by cognitive science findings on forgetting mechanisms. In addition, AKT devised a context-aware metric to quantify the time difference between exercises previously answered by students. AKT uses a set of embeddings derived from the Rasch model [62], a clear and interpretable IRT model [22], to capture individual variations between exercises without introducing extra model parameters.

3.3.3. Separated self-attention neural KT (SAINT) and (SAINT+)

Choi et al. [58] conclude that the attention layer of the SAKT model is too shallow and the computation of E, K, V lacks empirical support, and propose the SAINT model to solve these two problems. As shown in Fig. 6, SAINT consists of two main components, encoder and decoder. The input of the encoder is the question inlay matrix $\hat{E} = (e_1, e_2, \ldots, e_t)$, e is obtained by concatenating the question inlay and the position inlay, and the decoder takes the output of the encoder and the answer inlay matrix $\boldsymbol{A} = (a_1, a_2, \ldots, a_2)$ as the input. The encoder and decoder are both a combination of a multiheaded attention module and a front-blind neural network, with the difference that the decoder has 2 attention modules overlaid to further capture the complex relationship between questions and responses. The internal implementation of the coder is the same as that of SAKT, with the output of the decoder $\boldsymbol{F}_{\text{D}}$ passing through a Sigmoid-activated linear layer to obtain the final prediction. In SAINT, the coder is overlaid with four layers, and the experimental results show that the attention layer with multiple additions can

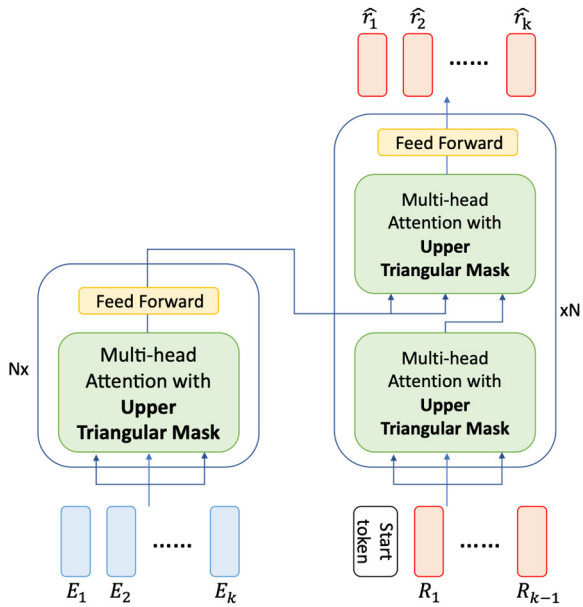

Fig. 6. An overview of SAINT.

effectively increase the area of AUC and achieve better prediction performance.

As a follow-up to SAINT, Shin et al. [60] suggest SAINT+, which is a Transformer-based knowledge tracing model that can handle practice information and student response information separately, as well as the lag time between responses. When it comes to practice embedding, the encoder applies a self-attention layer, while the decoder applies an encoder–decoder attention layer and, finally, to the response embedding stream, SAINT+ follows the SAINT structure.

3.3.4. Exercise-Enhanced RNN (EERNN) and exercise-aware KT (EKT)

Su et al. [36] uses cosine similarity to determine the attention weights between hidden states w_1.

$$w_t = \sum_{i=0}^{l} \cos\left(e_t, e_i\right) \cdot h_i. \tag{13}$$

Notably, the cosine similarity does not measure the similarity between the concealed states, but rather the similarity between the concepts.

$$\boldsymbol{r}_t = \mathrm{ReLU}\left(\boldsymbol{W}_1 \cdot \left[\boldsymbol{w}_{t+1} \oplus \boldsymbol{e}_{t+1}\right] + \boldsymbol{b}_1\right) \tag{14}$$

$$\boldsymbol{y}_t = \mathrm{Sigmoid}\left(\boldsymbol{W}_2 \cdot \boldsymbol{r}_t + \boldsymbol{b}_2\right). \tag{15}$$

Then Liu et al. [63] extend EERNN to an explainable Exercise-aware Knowledge Tracing (EKT) framework by incorporating the knowledge concept information, where the student's integrated state vector is now extended to a knowledge state matrix, in order to explicitly track their knowledge acquisition across multiple knowledge concepts. Each exercise may be quantified for how much it affects students' comprehension of numerous concepts while they are exercising in EKT. The EKT model combines the EERNN model with the DKVMN model, and thus has a dual attention module.

3.3.5. Relation-aware self-attention KT (RKT)

Pandey et al. [34] offer a relation-aware self-attention model for Knowledge Tracing (RKT) which adds a relation-aware contextual self-attention layer. This contextual information mixes textual exercise related information, student performance data, and forget behaviour information by modelling an exponentially declining kernel function. RKT combines attention weights with the connection coefficients of the exercise relation model and the forgetting behaviour model. It uses problem texts to represent problems and estimates their connections in past communication sequences for exercise-relationship modelling.

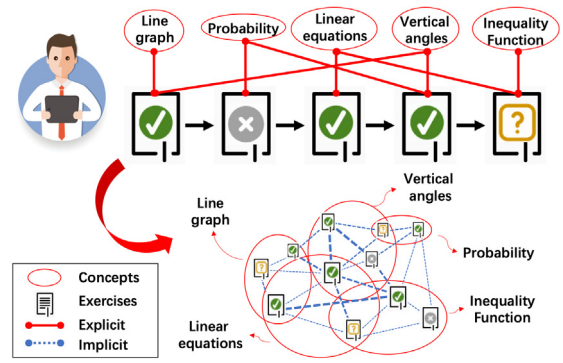

Fig. 7. An overview of Graph Structural Deep Learning-based KT.

3.4. Graph structural deep learning-based KT models

Various relationship patterns often appear in KT tasks, such as similarity of concepts, interdependence of concepts, and correlation between exercises and their concepts. Recent trends examine the ability of graph representation learning approaches, like as Graph Neural Networks (GNNs), to capture such relationship patterns in order to better handle the KT issue. This section provides a quick summary of the most important graph-based knowledge tracing models and discusses each one independently (see Fig. 7).

3.4.1. GNN-based knowledge tracing (GKT)

Nakagawa et al. [33] propose the GKT model, which transforms the knowledge tracing task into a time-series node-level classification problem in a graph neural network (GNN) by representing the relationship between KC as a directed graph. The directed graph $G = (V, E, A)$ consists of the set of nodes (denoted concept) $V = \{v_1, v_2, \ldots, v_N\}$, the set of edges (denoting the relationship between concepts) $E \subseteq V \times V$ and the adjacency matrix (defining the weight values of the relations) $A \in \mathbb{R}^{N \times N}$ defined. In GKT, assume that the student has an independent knowledge state $s^t = \left\{s^t_{i \in V}\right\}$ for each knowledge point v_i at each moment t. When students answer questions related to the concept v_i, the knowledge states s^t_i and $s_{j \in N_i}$ corresponding to v_i and the knowledge points adjacent to v_i are updated. Where, N_i denotes the set of knowledge points adjacent to v_i. Therefore, for a node v_i, first aggregate its knowledge states with the neighbouring nodes $j \in N_i$.

$$\boldsymbol{s}'^t_t = \begin{cases} \left[\boldsymbol{s}^t_i, \boldsymbol{a}^t \boldsymbol{E}_x\right], l = i, \\ \left[\boldsymbol{s}^t_l, \boldsymbol{E}_k(l)\right], l \neq i, \end{cases} \tag{16}$$

where $\boldsymbol{E}_x, \boldsymbol{E}_k$ are the interaction and KC embedding matrices respectively, and $\boldsymbol{E}_k(l)$ represents the łth row of $\boldsymbol{E}_k$. Update the knowledge state according to the graph structure:

$$\hat{m}^{t+1}_i = G_{\mathrm{ea}}\left(\boldsymbol{m}^{i+1}_t\right), \boldsymbol{s}^{t+1}_l = G_{\mathrm{GRU}}\left(\hat{\boldsymbol{m}}^{t+1}_l, \boldsymbol{s}^t_l\right). \tag{17}$$

Where, f_{self} is a multilayer perceptron, G_{ea} is a division-add mechanism in the DKVMN model, G_{GRU} is the GRU model, and f_{neighbor} is a function based on the graph structure information function that propagates to neighbouring nodes based on graph structure information.

3.4.2. Graph-based interaction model for KT (GIKT)

Yang et al. [64] suggest a graph-based knowledge tracing model (GIKT) to connect exercise information with higher-order exercise-concepts. GIKT uses graph convolutional networks (GCNs) to encode exercise and concept relationships. GIKT generalises students' mastery of exercises as an interaction between their present condition, relevant practice history, target problems, and relevant abilities.

GIKT generalises the following parts of the interaction: $\langle \mathbf{h}_t, \widetilde{\mathbf{e}}_t \rangle$ is used to indicate the student's level of exercise mastery e_t, $\langle \mathbf{h}_t, \widetilde{\mathbf{s}}_j \rangle$ to reflect the student's degree of mastery of the associated concept; GIKT also expands the interaction on present student state to historical interactions that demonstrate the relevant history competence, i.e., $\langle \mathbf{e}_i, \widetilde{\mathbf{e}}_t \rangle$ and $\langle \mathbf{e}_i, \widetilde{\mathbf{s}}_j \rangle$, $\mathbf{e}_i \in \mathcal{I}_e$.

Then, GIKT predicts based on all of the above interactions and define the generalised interaction module. To encourage relevant interactions and cut down on noise, GIKT uses an attention network to learn bi-attention weights for all interaction terms and then use the weighted sum as the prediction:

$$\begin{aligned} \alpha_{i,j} &= \mathrm{Softmax}_{i,j}\left(\boldsymbol{W}^T\left[\boldsymbol{f}_i, \boldsymbol{f}_j\right] + b\right) \\ p_t &= \sum_{\boldsymbol{f}_i \in \mathbf{I}_e \cup \{\mathbf{h}_t\}} \sum_{\boldsymbol{f}_j \in \widetilde{\mathbf{N}}_{q_t} \cup \{\widetilde{\mathbf{q}}_t\}} \alpha_{i,j} g\left(\boldsymbol{f}_i, \boldsymbol{f}_j\right) \end{aligned} \tag{18}$$

3.4.3. Pre-training question embeddings (PEBG)

Liu et al. [65] show that considerable improvements in KT may be acquired by pre-training embeddings for each exercise using rich side information and then training deep learning based KT models using the obtained embeddings. Specifically, the side information comprises the exercise's complexity and the relationships between the exercise and the concept included in the bipartite graph.

There are edges between exercise vertices and concept vertices in the exercise-concept bipartite network, representing an explicit signal. Similar to the modelling of 1st-order proximity in LINE [66], explicit relationships are modelled by taking into account the local closeness of concept and exercise vertices. Specifically, we utilise inner products to estimate the local closeness of exercise and concept vertices in the embedding space:

$$\hat{r}_{ij} = \sigma\left(\boldsymbol{q}_i^T \boldsymbol{s}_j\right), i \in [1, \ldots, |Q|], j \in [1, \ldots, |S|] \tag{19}$$

where $\sigma(x) = 1/\left(1 + e^{-x}\right)$ is the sigmoid function, which transforms the relation value to a probability.

3.4.4. Joint graph convolutional network based KT (JKT) and Bi-graph contrastive learning based KT (Bi-CLKT)

Song et al. [37] present a joint graph convolutional network based deep Knowledge Tracing (JKT) framework to express multi-dimensional relationships between "exercise-to-exercise" and "concept-to-concept" in a graph and combine them with "exercise-to-exercise" interactions. JKT makes it possible to link exercises that target diverse ideas, as well as to collect and make the model's high-level semantic data more comprehensible (see Fig. 8).

Given an exercise set $\mathcal{V}_e$ and an "exercise-to-exercise" interaction set $\mathcal{E}_e$, an exercise influence sub-graph is a graph $\mathcal{G}_e = (\mathcal{V}_e, \mathcal{E}_e)$ that takes the exercises in $\mathcal{V}_e$ as the vertices and the interactions in $\mathcal{E}_e$ as the edges. The weighted influence $Q_e^{ij}(i, j \in \mathbb{N}, i, j \leqslant m)$ of the edges of co-occurred exercises $\mathcal{V}_e^i$ and $\mathcal{V}_e^j$ is measured by

$$Q_e^{ij} = \frac{f^c\left(\mathcal{V}_e^i, \mathcal{V}_e^j\right)}{\sum_m f^o\left(\mathcal{V}_e^i, \mathcal{V}_e^m\right)}$$

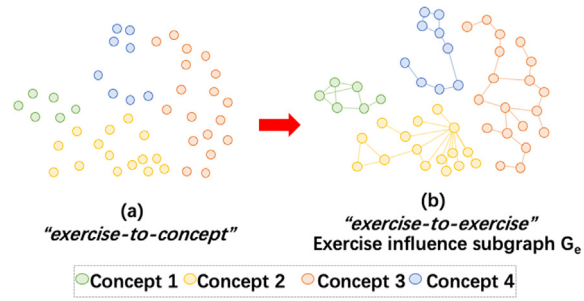

Fig. 8. An overview of an exercise influence sub-graph.

where Q_e^{ij} is the co-correctness rate of exercises $\mathcal{V}_e^j$ and $\mathcal{V}_e^i$ among all the answered co-occurred exercises involving $\mathcal{V}_e^i \cdot f^c$ and f^0 denote the count of co-correctness and co-occurrence respectively. Edge $\mathcal{E}_e\left(\mathcal{V}_e^j, \mathcal{V}_e^i\right)$ only exists when Q_e^{ij} is larger than a certain threshold τ.

Song et al. [67] further explore techniques to combine exercises with rich structural information and first propose Bi-Graph Contrastive Learning-based Knowledge Tracing, which is inspired by this concept and contemporary developments in self-supervised learning (Bi-CLKT). They devise a two-level contrastive learning technique based on a "exercise to exercise" (E2E) concept. It involves node-level contrastive learning of subgraphs for the production of discriminative representations of exercises and graph-level contrastive learning for the acquisition of discriminative representations of ideas.

4. Deep learning-based KT models comparison & analysis

It is a well-known fact that deep learning improves the effectiveness of knowledge tracing. For the rapidly expanding field of online education, the vast quantity of educational data created meets the data volume requirements of deep learning models. The absence of labelled data facilitates the usage of data, while the prevalence of deep learning frameworks facilitates the construction of models. Due to all of these factors, Deep Learning-based KT is popular on online education platforms and a hotbed of educational data mining research.

In Section 3, improvement strategies for the DKT model are presented in depth. The approaches are categorised according to the sort of enhancement they provide: enhancements for explainable issues, enhancements for dependent problems, the addition of features, and additional enhancements. This section provides an overview of the datasets used by each kind of optimisation technique. The performance of Deep Learning-based KT on each public dataset is then analysed and compared. This section covers the datasets used for each technique of improvement and compares and ranks Deep Learning-based KT's performance on a variety of public datasets.

4.1. Datasets

There are six available datasets to support the research of knowledge tracing. Their statistic information has been listed in Table 2.

- ASSISTments2009 (ASSIST 2009)[1]: This 2009–2010 dataset includes 525,535 interactions (student answers), including duplicates. This dataset includes 346,860 student interactions with 26,688 exercises and 123 concepts in which

[1] https://sites.google.com/site/assistmentsdata/home/assistment-2009-2010-data

17,751 exercises have concepts. exercises without concepts are marked 'NA' or null, while some have four. This dataset's initial version is unreliable, as described. The updated version 'skill-builder'3 fixes data modelling problems and removes duplicates.

- ASSISTments2012 (ASSIST 2012)[2]: This is the version of the ASSISTments datasets with the most information. It has information that was gathered over the course of a year. Even though the ASSISTments team claims the dataset has about 10 million "exercises", the real dataset has 179,999 unique exercises that 46,674 students answered, leading to 6,123,270 interactions. We can see that most exercises (126,908) do not have any of the linked concepts.
- ASSISTments2015 (ASSIST 2015)[3]: This dataset demonstrates how 708,631 students using ASSISTments in 2015, while 19,917 students responded to 100 exercises. The dataset has four characteristics: the question IDs, the identities of the students who answered the exercises, an attribute indicating the accuracy of each student's response, and a log attribute indicating how the responses were provided over time. Unlike previous ASSISTments datasets, no metadata nor concepts are included. The average number of responses to each question also varies. Each answer in the 2009 and 2012 versions has 12.99 and 34.01 on average respectively.
- ASSISTment Challenge (ASSISTChall)[4]: The 2017 release of the complete ASSISTment Challenge dataset includes information from the academic years 2004–2005 and 2005–2006. It consists of 3,162 unique exercises answered by 1,709 students over 102 concepts, yielding 942,820 interactions. This dataset ranks second in terms of answer per exercise ratio, with an average of 298.17 per response (only lower than the ASSISTments2015 dataset).
- STATICS2011 Dataset[5]: This dataset includes student responses to exercises from the Fall 2011 Engineering Statics course at Carnegie Mellon University (CMU). 361,092 interactions, 335 students, 1,224 exercises compose the original dataset. Commonly, this dataset is preprocessed, yielding in 1,224 exercises answered by 333 students and 85 concepts. Preprocessing decreases the number of student interactions to 189,297. The large number of interactions without exercise-answer information supported this preprocessing. During preprocessing, 'issue name' and 'step name' are concatenated and only legitimate first attempts are considered. This dataset averages 568.45 answers per exercise.
- Simulated-5 Dataset[6]: Piech et al. [25] proposed a synthetic dataset that mimics the responses of virtual students in a controlled environment. There are training and testing subsets in the dataset. Each subset consists of 50 exercises with a concept and degree of difficulty. 4,000 virtual students respond to each exercise, accumulating two hundred thousand interactions. To imitate student learning over time, item response theory was used. This dataset has a uniform format and does not need any preparation, such as removing duplicates or inferring a student's exercise sequence, allowing for direct comparison of KT models.

[2] sites.google.com/site/assistmentsdata/home/2012-13-school-data-with-affect

[3] sites.google.com/site/assistmentsdata/home/2015-assistments-skill-builder-data

[4] sites.google.com/view/assistmentsdatamining

[5] https://pslcdatashop.web.cmu.edu/DatasetInfo?datasetId=507

[6] https://github.com/chrispiech/DeepKnowledgeTracing/tree/master/data/synthetic

Table 2
Statistic information of datasets.

Dataset	#Exercises	#Students	#Interactions	#Concepts
ASSISTments2009	26,688	4217	346,860	123
ASSISTments2012	179,999	46,674	6, 123, 270	265
ASSISTments2015	100	19,917	708,631	–
ASSIST Chall	3162	1709	942,816	102
STATICS2011	1224	335	361,092	85
Simulated-5	50	4000	200,000	5

4.2. Evaluation metrics

In KT tasks, most of the methods use binary classification to predict student performance, such as whether an exercise is completed correctly or not. Therefore, usually we evaluate performance by comparing the area under the curve (AUC) metric. In the training phase, we train the model with interactions, and in the testing phase, we update the model after receiving the answers for each exercise. After this, the improved model was used to provide predictions for future exercises. The area under the curve (AUC) value of 0.5 shows the performance prediction results based on random guesses; the larger it is, the better it is.

4.3. Experimental results and discussion

Table 3 compares the performance of KT models in predicting students' future responses on all datasets. We provide the AUC intervals for each model (based on data taken from previous significant papers). However, due to some variability in the settings and parameters taken by each model as a comparison model across papers, these values may appear to span a wide range. We try to provide some necessary analysis as well as conclusions in these data.

For example, on the ASSISTments dataset, Bi-CLKT outperforms most of the other KT algorithms due to the new structural model it introduces, which contributes to the accuracy of the localisation of each exercise; however, on the ASSISTment 2009 dataset, DKT (the highest value) outperforms Bi-CLKT, and here there are some data imbalances on the dataset balance problem. The new structural model, such as the topology of the figure, is more flexible than the recurrent neural network and, therefore, better captures the rich information in the large-scale real-world student response dataset. On average, the models with the new structure perform better on larger datasets. This suggests that the new structural models, such as the topology of the figure, are more adaptive than recurrent neural networks. The models with the added attention mechanism had about 3% higher AUC in the ASSISTments2015 and ASSISTments challenge datasets compared to the other datasets. When applied to the Statics2011 and Simulated-5 datasets, most of the models performed at their maximum capacity.

There are some differences in the performance of these models on each of these mainstream datasets, however, with the development of deep learning, we can see that more and more new structures play a positive role in facing the problem of KT, and they also improve some limitations and problems of the established models, for example, the lack of interpretability.

5. Conclusion and future research directions

More and more student interactions are taking place on online education platforms, which has facilitated the rapid development of AI-assisted educational systems in the era of the big data boom and pandemic. The expansion of student interaction data and the rapid proliferation of real-world application problems necessitate advancements in cognitive judgement and mastery

Table 3
Experimental results (AUC).

Model	AUC Interval					
	ASSIST 2009	ASSIST 2012	ASSIST 2015	ASSIST Chall	Simulated-5	STATICS2011
DKT [25]	0.709∼0.86	0.712∼0.731	0.707∼0.737	0.726∼0.734	0.75∼0.826	0.815∼0.823
DKT+ [14]	0.802∼0.823	–	0.728∼0.737	0.712∼0.734	0.824∼0.826	0.830∼0.835
DKT-DSC [55]	0.735	0.721	0.716	0.43	–	–
EERNN [36]	0.735	0.721	0.716	0.43	–	–
DKVMN [35]	0.8157	–	0.7268	–	0.827	0.8284
SKVMN [57]	0.8363	–	0.7484	–	0.840	0.8485
SAKT [34]	0.686∼0.848	0.691∼0.735	0.721∼0.854	0.657∼0.734	0.832	0.803∼0.853
AKT [58]	0.747∼0.835	0.7555	0.7828	0.7702	–	–
SAINT [58]	–	–	–	–	–	–
SAINT+ [60]	0.782	–	0.754	0.734	–	–
RKT [34]	–	0.793	–	–	–	–
GKT [33]	0.723	0.735	–	–	–	–
GIKT [64]	0.790	0.775	–	–	–	–
PEBG [65]	0.83	0.77	–	–	–	–
JKT [37]	0.798	–	0.765	0.745	–	0.856
Bi-CLKT [67]	0.857	–	0.765	0.775	–	0.865

strategies for students in order to enhance the overall efficacy of the educational system. As a result, AI-assisted educational applications, such as deep learning-based KT, have garnered increased attention.

This study summaries knowledge tracing methodologies and strategies based on deep learning for big educational datasets. To provide a clear roadmap for readers interested in studying deep learning-based KT techniques, we provide a summary of these methods' technological mechanisms, important improvements, and major technical contributions. Based on the composition of each KT approach, we categorise them into four technical mechanisms: RNN model-based DKT and its variants, memory network-based, attention mechanism-based, and graph structural-based. In addition to the broad concepts and procedures of each method, we highlight the originality of the state-of-the-art approaches that prompted researchers to establish their techniques within the subsection of each method. In addition, we summary the strengths and shortcomings of each model and outline the knowledge-tracing strategies based on deep learning for big educational datasets, as well as the contributions of each methodology in three major improvement categories. In Section 4, we give the six most widely used open datasets within the KT area to facilitate and encourage future researchers. In addition, we present the most frequent evaluation measures for KT. Then, we compare and evaluate the outcomes of several available benchmark tests derived from the examined KT research.

Despite the fact that Deep learning-based KT was initially proposed 20 years ago, there are no criteria for determining which Deep learning-based KT strategy is the best, as each technical approach has its own pros and disadvantages. Briefly, deep learning has not yet solved the interpretability challenge. The self-attentive paradigm handles long-term dependencies flawlessly, although positional encoding is necessary to preserve sequence learning capabilities. Deep learning-based KT is dominated by research on missing learning features, and each of the three approaches – embedding, loss function restriction, and new structures – has advantages and disadvantages: embedding is intuitive but overly dependent on the learning ability of the model, loss function restriction allows specifying the direction of model optimisation but requires significant manual effort, and the use of new structures offers many advantages but lacks inference.

Open Issues: In addition to the three major improvement issues listed in Table 1, the other potential challenging problems should be paid more attention to:

- **Limitations of binary problem:** To the best of our knowledge, the problems encountered by students in the real world are multiple and complex, yet the vast majority of KT models based on deep learning at this stage explain binary problems rather than problems involving subjective scoring. Therefore, the processing of higher complexity exercises is still an unexplored area at this stage.
- **Introduction of new learning features:** It is difficult to introduce additional learning features based on learning trajectory data because deep learning-based KT is widely utilised in online education platforms. It is difficult to implement new features. Length of the input vector has an effect on the training speed of KT for RNN-based deep learning. Reducing the length of a vector requires dimension reduction or embedding techniques). Users must extract, model, and provide feature data to the application using KT based on deep learning.
- **Integration of knowledge graphs:** Models of KT that include deep learning are able to identify knowledge point relationships and generate a knowledge graph. Improving the model's capacity to identify knowledge structures and expanding the knowledge point connection graph to a knowledge graph will be crucial future research topics.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

This work was mainly supported by the ARC Linkage Project under Grant No. LP180100750; ARC Discovery Projects under Grant No. DP200103700, DP220102191 and Study Melbourne Research Partnership program under Project ID: veski-SMRP#1906.

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